HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 2

CANDIDATE NAME		**CT GROUP**	**17S7___**
CENTRE NUMBER		**INDEX NUMBER**	

BIOLOGY — 9744/01

Paper 1 Multiple Choice Questions — **20 September 2018**

Additional Materials: Optical Mark Sheet — **1 hour**

INSTRUCTIONS TO CANDIDATES

1. Write your **name** and **CT group** in the spaces provided at the top of this cover page.
2. Fill in your particulars on the Optical Mark Sheet. Write your **NRIC number** and shade accordingly.
3. There are **thirty** questions in this paper. Answer **all** questions. For each question, there are four possible answers, **A**, **B**, **C** and **D**.

   Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.
4. At the end of the paper, you are to submit **only** the Optical Mark Sheet.

INFORMATION FOR CANDIDATES

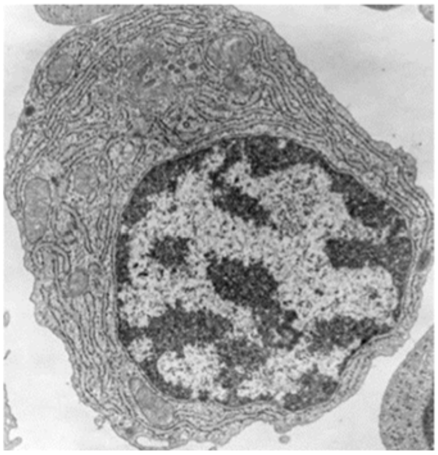
Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

The used of an approved scientific calculator is expected, where appropriate.

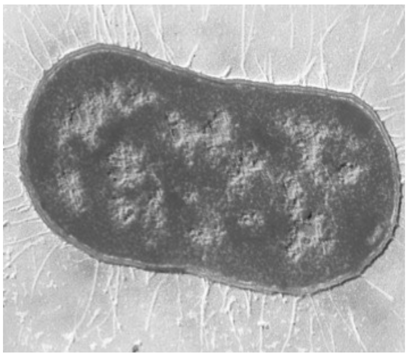
This document consists of **28** printed pages.

1 The electron micrographs 1, 2 and 3 show three different types of cells.



1

2

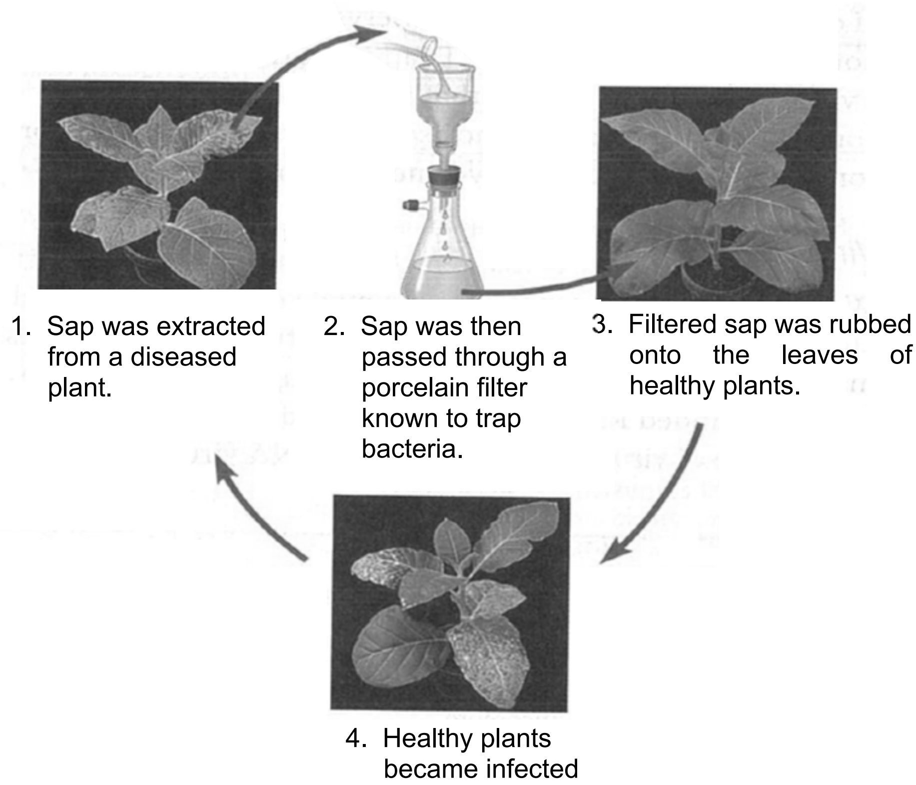


3

Which row is correct?

	electron micrograph of cell type	organelle found in cell type	description of organelle
A	1, 2 and 3	ribosomes	embedded on the smooth endoplasmic reticulum for synthesis of polypeptides
B	1 and 2 only	vacuoles	membrane-bound fluid-filled sacs
C	3 only	nucleoli	dense masses within the nuclei
D	2 only	centrioles	found in animal cells but absent in plant cells

2 An investigation into the properties of an agent that causes a plant disease was carried out.



The process was repeated several times and each successive group of plants developed the disease to the same extent as earlier groups.

Which statements, supporting the idea that the infectious agent is a virus, can be concluded from the experiment?

1 Viruses cannot replicate outside of a host cell.

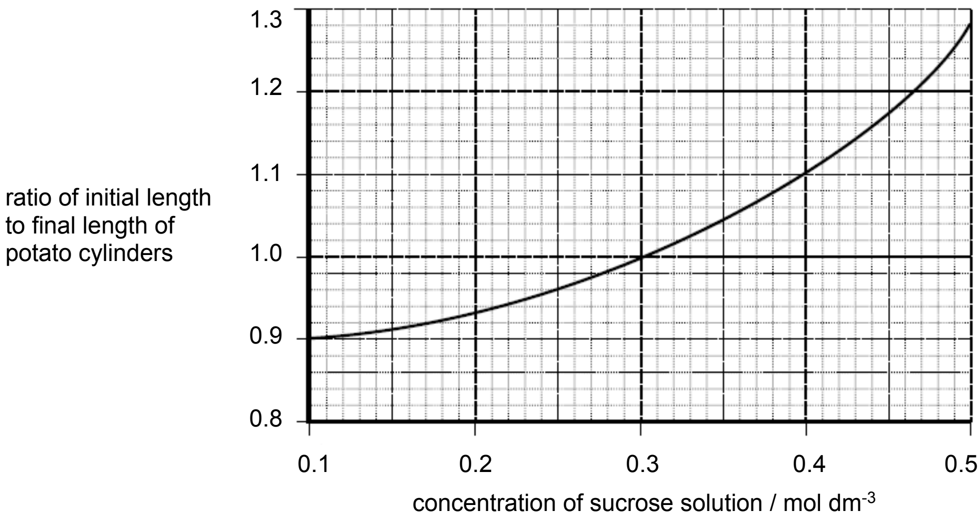
2 Viruses are smaller than bacterial cells and can pass through a bacterium-trapping filter.

3 Viruses can replicate within plant cells, so their ability to cause disease remains undiluted after several transfers from one healthy plant to another.

A 1 only **B** 1 and 2 **C** 2 and 3 **D** 1, 2 and 3

3 In an experiment, potato cylinders were placed in sucrose solutions of different concentrations. The potato cylinders were measured before and after immersion into each sucrose solution.

The graph shows the effect of different sucrose concentrations on the length of the potato cylinders.



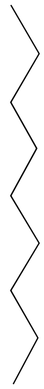
When the sucrose solution was less than 0.3 mol dm^{-3}, water molecules moved from a region of …1… negative water potential to a region of …2… negative water potential. Hence, water molecules moved from the ...3… into the …4….

If the initial length of the potato cylinder in 0.1 mol dm^{-3} sucrose solution is 5.0 cm, the final length will be …5…cm.

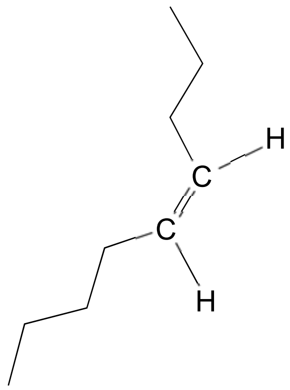
Which words correctly fill gaps 1, 2, 3, 4 and 5?

	1	2	3	4	5
A	less	more	sucrose solution	potato cells	5.5
B	less	more	sucrose solution	potato cells	4.5
C	more	less	potato cells	sucrose solution	5.5
D	more	less	sucrose solution	potato cells	4.5

4 Fatty acid chains of natural membrane phospholipids may be saturated or unsaturated.



fatty acid chain A



fatty acid chain B

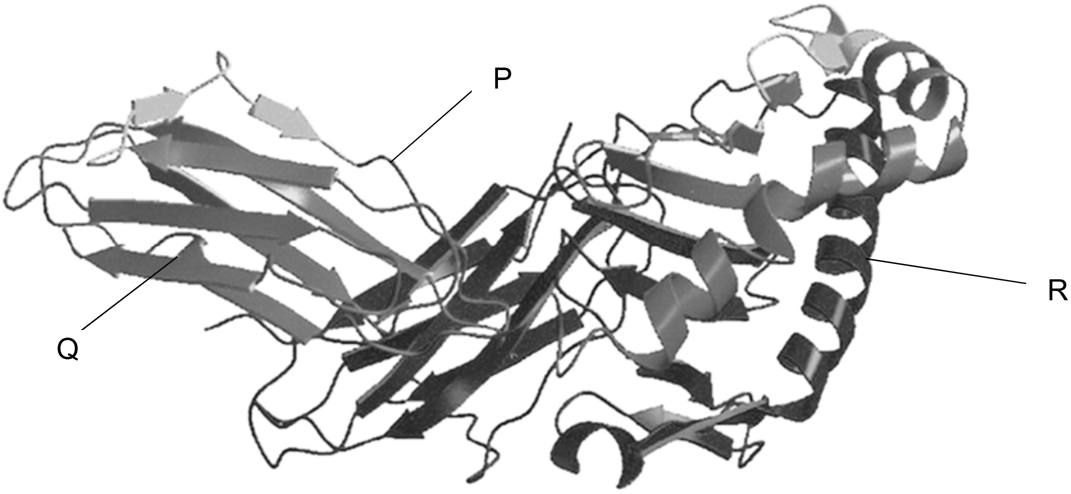
Which statements regarding fatty acid chains A and B are correct?

1. The presence of phospholipids with mainly fatty acid chain A in cell membranes have increased membrane viscosity.
2. Fatty acid chain B is unsaturated because it contains one double bond creating a kink in the fatty acid structure.
3. A membrane containing predominantly fatty acid chain B has a greater fluidity than one which predominantly contains fatty acid chain A.
4. The presence of phospholipids with fatty acid chain B eases the passage of small, non-polar molecules, such as oxygen through membranes.

A 1, 2, 3 and 4 **B** 1, 3 and 4 only **C** 1 and 2 only **D** 2 and 4 only

5 The diagram shows the ribbon model of the cell surface membrane receptor, HLA-DR1.

HLA-DR1 consists of two polypeptides, where parts of the protein are labelled P, Q and R.



Which statement regarding the cell surface membrane receptor HLA-DR1 is correct?

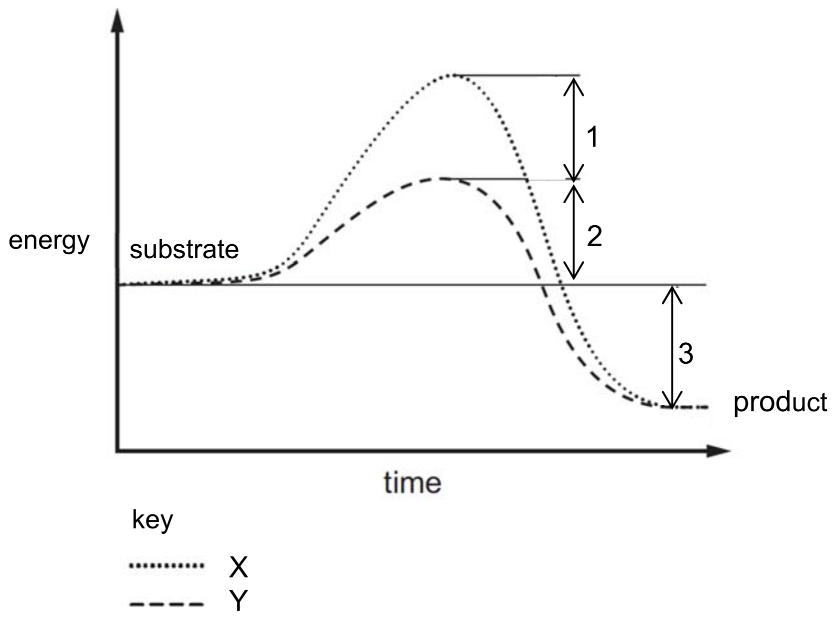
A In the formation of structure P, each hydrogen bond is formed between the –NH of one amino acid and the C=O group of another amino acid.

B Q is important for the folding of the polypeptide into a specific 3D conformation of its active site.

C R-group hydrogen bonding is involved in the formation of structure R.

D Structures P, Q and R play a role in the formation of the quaternary structure of HLA-DR1.

6 The graph shows the effect of an enzyme on a reaction.



Which row correctly identifies the labels?

	reaction with enzyme	difference in activation energy	energy released during reaction
A	X	1	2 + 3
B	Y	1 + 2	3
C	X	1 + 2	1 + 2 + 3
D	Y	1	3

7 Once most stem cells differentiate, they lose their ability to turn into other types of cells. However, some fully differentiated cells can be stimulated to change back into stem cells in tissue culture. Such cells are called induced pluripotent stem cells (iPS cells).

In experiments with mice, it was discovered that the introduction of four genes would cause certain fully differentiated cells to change to iPS cells. Genes were introduced into host mouse cells using artificially synthesised plasmids.

There is evidence to suggest that the introduction of the four genes caused an increase in the production of telomerase reverse transcriptase (TERT) in the fully differentiated cells.

Which statements are correct?

1 The iPS cells are useful for research because they have the ability to differentiate into cells of the extra-embryonic membranes.

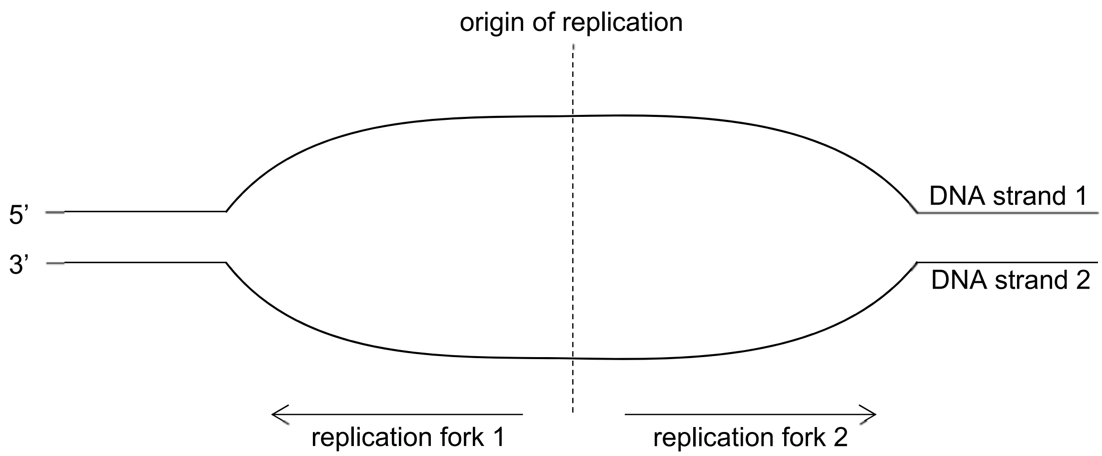
2 Only a few genes were required to be added because these genes were likely able to influence the activity of many other genes by coding for transcription factors.

3 TERT, which plays a part in maintaining the length of the telomere at the end of a chromosome, is not normally switched on in differentiated cells.

4 This method of creating iPS cells has less ethical concerns than harvesting embryonic stem cells.

A 1, 2 and 3 **B** 1, 2 and 4 **C** 1, 3 and 4 **D** 2, 3 and 4

8 The diagram shows the replication bubble of part of a DNA molecule.

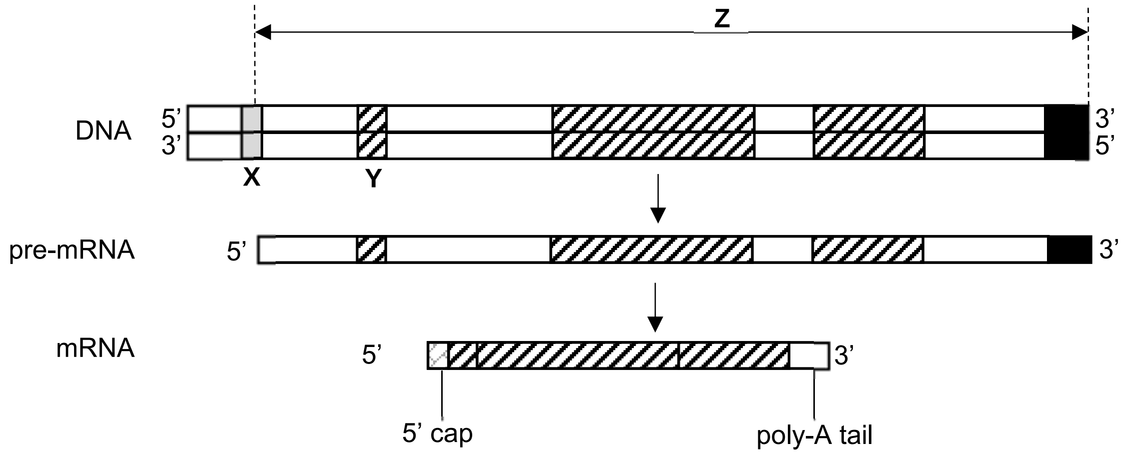


Which statements regarding the replication of DNA are correct?

1 At replication fork 1, synthesis of the complementary strand of DNA strand 1 is continuous.
2 At replication fork 2, DNA strand 1 requires multiple primers for the replication process.
3 Four DNA molecules are newly formed upon the completion of replication.

A 1, 2 and 3 **B** 2 and 3 **C** 1 and 2 **D** 1 only

9 The diagram shows the transcription of a gene in a eukaryotic cell.



Which statements are correct?

1 X is not transcribed except for the transcription start site.
2 RNA polymerase and transcription factors bind to Y to initiate transcription.
3 Z comprises the promoter, coding region and termination sequence.

A 1, 2 and 3

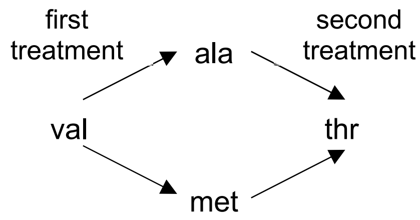
B 1 and 3 only

C 2 and 3 only

D 1 only

10 A bacteria colony produces a normal protein with a known amino acid sequence.

The bacteria colony was treated with the same chemical mutagen twice. This gave rise to two mutant strains of bacteria where each had a single nucleotide change at a particular mRNA codon resulting in a change of amino acid as shown in the diagram.

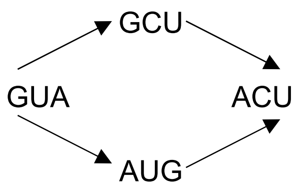


The mRNA codons for some amino acids are shown in the table.

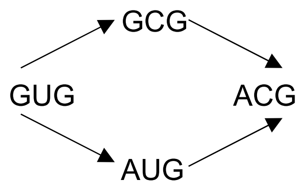
alanine (ala)	leucine (leu)	threonine (thr)	valine (val)
GCU GCC GCA GCG	UUU UUC UUA UUG	ACU ACC ACA ACG	GUU GUC GUA GUG

Assuming that both treatments resulted in a single nucleotide change each, which diagram correctly shows the codons that were translated into alanine, methionine, threonine and valine?

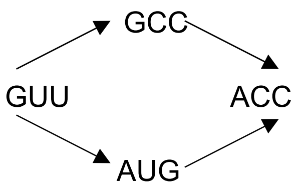
A



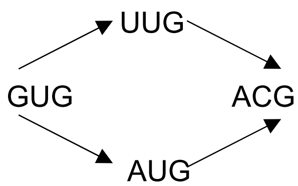
B



C



D



11 In order to replicate, the ends of a eukaryotic chromosome contain a special sequence of DNA called a telomere.

When cells undergo mitotic division, some of these repeating sequences are lost. This results in the shortening of the telomeric DNA.

What is a consequence of the loss of repeating DNA sequences from the telomeres?

A The cell will begin the synthesis of different proteins.

B The cell will begin to differentiate as a result of the altered DNA.

C The number of mitotic divisions the cell can make will be limited.

D The production of mRNA will be reduced.

12 Gene transfer refers to the movement of genetic material between cells. There are two types of gene transfer in prokaryotes:

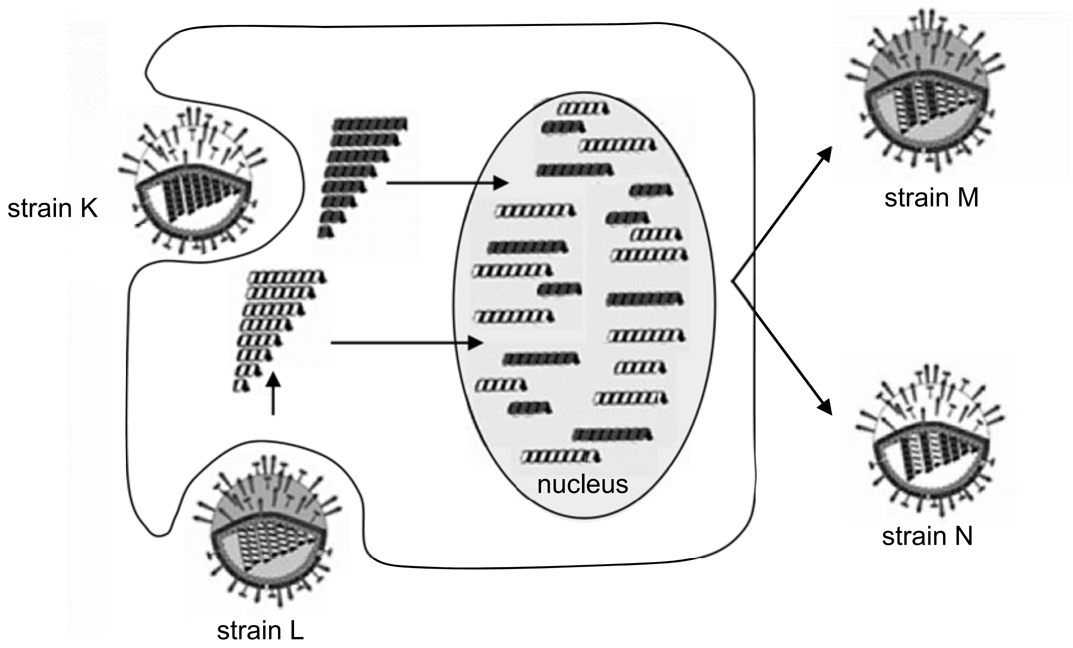
- vertical gene transfer where the transfer of genes is from parent to offspring
- horizontal gene transfer where the transfer of genes is between two bacterial strains.

Which statements correctly describe the two types of gene transfer?

1 Horizontal gene transfer consists of binary fission, conjugation and transduction.
2 Horizontal gene transfer and not vertical gene transfer results in genetic variation.
3 Both forms of gene transfer lead to bacterial strains acquiring new antibiotic resistance.

A 1, 2 and 3 **B** 1 and 3 **C** 2 only **D** 3 only

13 The diagram show the emergence of new influenza strains, M and N, resulting in antigenic shift in influenza A.

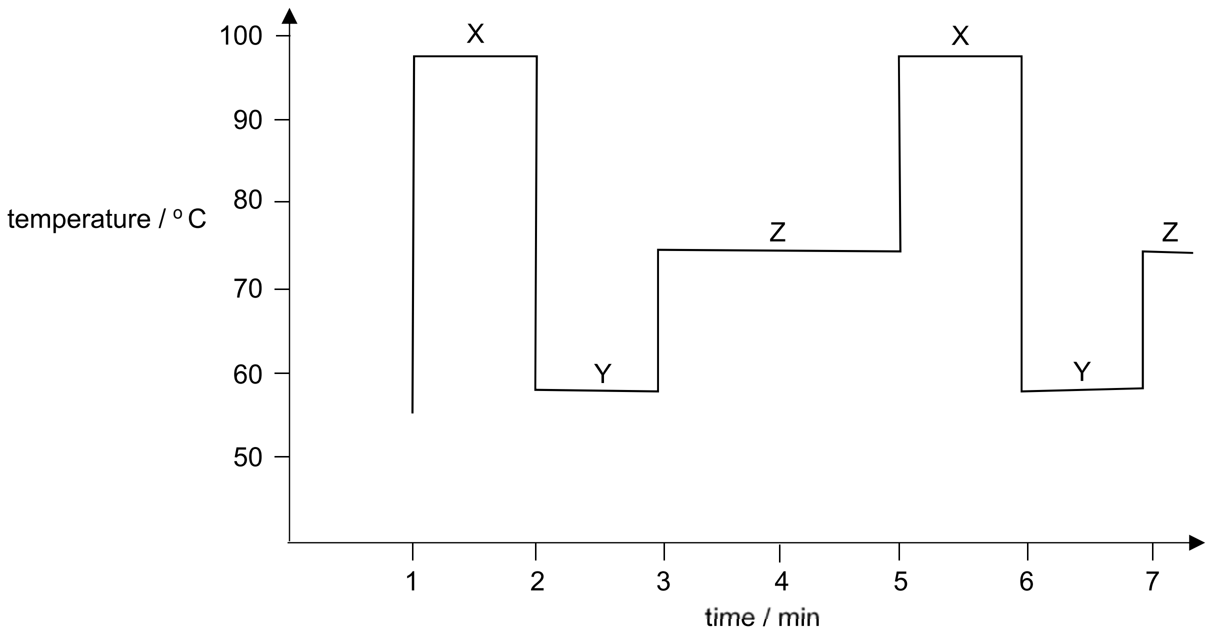


Which events lead to antigenic shift?

	name of event	description
1	mutation	mutations in the viral genomes of strains K and L occurred in the cell resulting in strains M and N
2	recombination of viral genome	recombination of strains K and L resulted in new strains M and N
3	reassortment of viral genome	reassortment of the viral genomes of strains K and L occurred only when both strains infect the same cell

A 1, 2 and 3 **B** 2 and 3 **C** 3 only **D** 2 only

14 The graph shows the thermal cycling of a DNA sample during polymerase chain reaction (PCR).

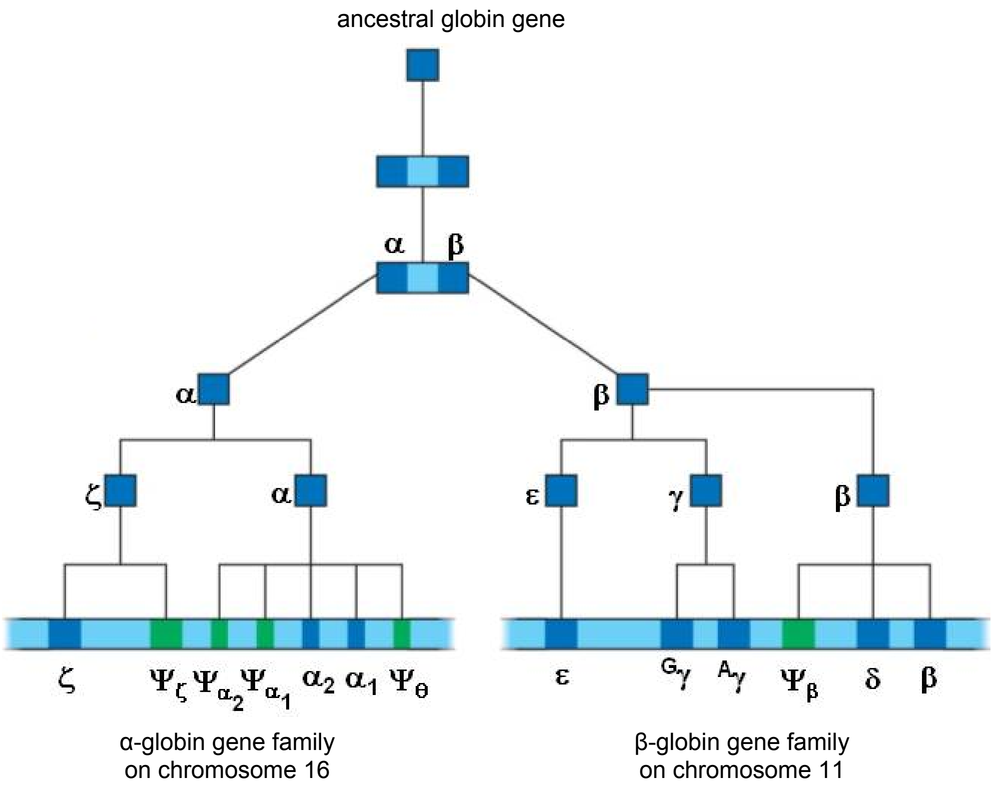


Which row is correct regarding the formation and breakage of bonds involved in PCR at steps X, Y and Z?

	step	stage in PCR	hydrogen bonds	phosphodiester bonds
A	Y	annealing of primers	forms	forms
B	Y	denaturation of DNA	forms	breaks
C	Z	extension of DNA	forms	forms
D	X	annealing of primers	breaks	breaks

15 Multigene families are defined as groups of genes with sequence homology and related overlapping functions. It is thought that these genes arose from a common ancestral gene that has accumulated mutations over time.

The globin gene family is one such example and the diagram shows how the gene family arose.



Which mechanism best explains how the globin gene family was formed?

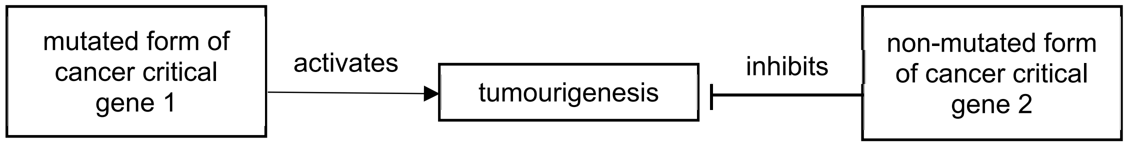
A Multiple unequal crossovers of the α and β globin RNA occur during mitosis to result in the different gene families.

B A germline mutation occurred where the ancestral globin gene was duplicated forming α and β globin gene, each with their own subsequent point mutations.

C A point mutation in the ancestral gene promoter resulting in the formation of a new origin of replication, and hence the repeated replication of a specific portion of the genome.

D Several copies of the α and β globin genes are duplicated over time, each with different degree of nonsense mutation leading to globin chains of varying functions.

16 Proto-oncogenes and tumour suppressor genes are cancer critical genes. The diagram shows how mutations in these genes play a role in the generation of tumours, which is known as tumourigenesis.



MicroRNA (miRNA) is a type of RNA that binds by complementary base pairing to mRNA. A miRNA, specific to the mRNA of either the mutated form of cancer critical gene 1 or the non-mutated form of cancer critical gene 2, is added to normal cells. The binding results in the repression of translation.

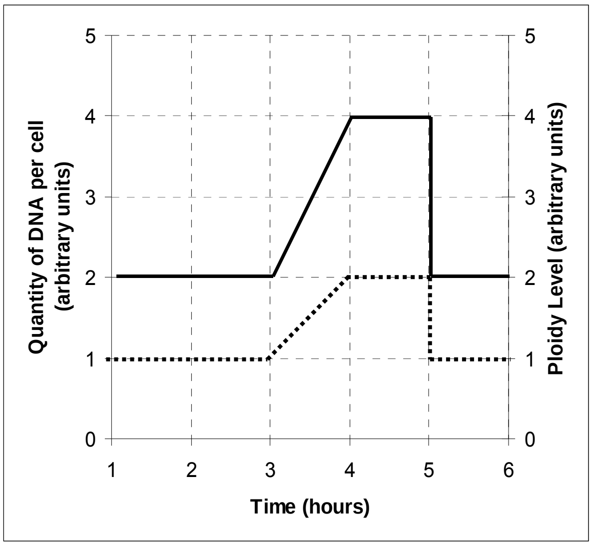
Which row is correct?

	tumour suppressor gene	binding of miRNA	result of miRNA binding
A	mutated form of cancer critical gene 1	mRNA of non-mutated form of cancer critical gene 2	tumourigenesis occurs
B	non-mutated form of cancer critical gene 2	mRNA of mutated form of cancer critical gene 1	no tumourigenesis
C	mutated form of cancer critical gene 1	mRNA of mutated form of cancer critical gene 1	tumourigenesis occurs
D	non-mutated form of cancer critical gene 2	mRNA of non-mutated form of cancer critical gene 2	no tumourigenesis

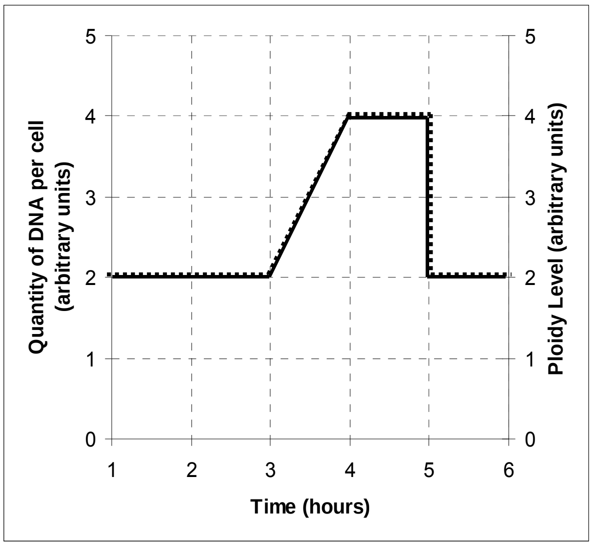
17 The ploidy level and quantity of DNA of a diploid cell varies during a cell cycle.

Which graph represents this variation?

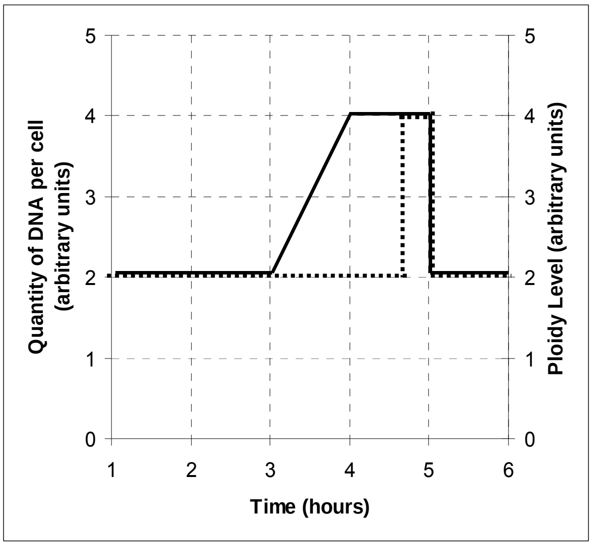
A



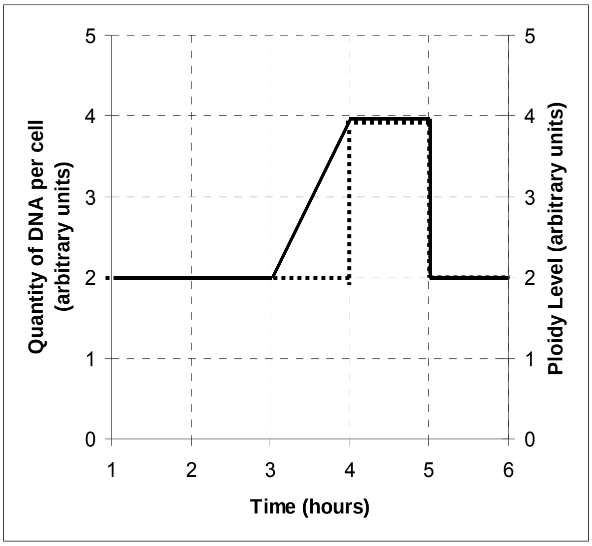
B



C



D

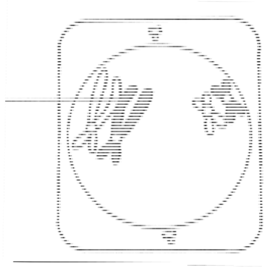


key

——— quantity of DNA

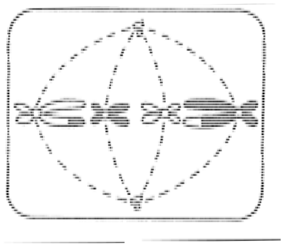
.... ploidy level

18 The diagram shows a cell undergoing meiosis.



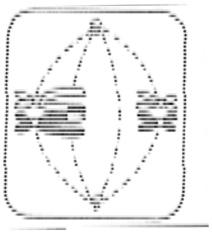
Which diagram represents the next stage in the process?

A

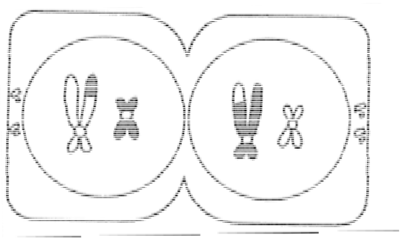


B

C



D



19 Read the following statements.

- Gibberellins belong to a group of chemicals known as terpenoids and are made up of the elements carbon, hydrogen and oxygen only.
- In the pea plants that Mendel studied, the stem length gene, *Le* / *le*, controls the length between nodes.
- Pure-breeding, tall pea plants were crossed with pure-breeding, dwarf pea plants. The F1 generation plants were all of the same height. When these were crossed, the numbers of tall and dwarf plants in the F2 generation were counted. There were 787 tall and 277 dwarf plants.
- The dwarf variety of the pea plant lacks gibberellin.
- Addition of gibberellin to the dwarf plants results in conversion of the dwarf to the tall phenotype.

What can be concluded from these statements?

A Heterozygous genotypes for the stem length gene are of an intermediate height, as only 50% of the product of gene expression is synthesised.

B The lack of a ratio of 3 tall pea plants to 1 dwarf pea plant in the F2 generation means that there is an environmental effect contributing to height in pea plants.

C The *Le* / *le* gene codes for the protein gibberellin, with only the *LeLe* or *Lele* genotypes expressing gibberellin and with the *lele* genotype unable to express gibberellin.

D There is at least one altered triplet code in the dwarf allele of the *Le* / *le* gene, producing a polypeptide with an altered tertiary structure and resulting in a non-functioning protein.

20 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

A one gene with a pair of co-dominant alleles

B one gene with multiple alleles

C sex linkage of the allele for grey coat colour

D two genes, each with a dominant and recessive allele

21 In the fruit fly, *Drosophila melanogaster*, four genes whose recessive alleles code for black body (B/b), curved wings (C/c), purple eyes (P/p) and vestigial wings (V/v) are linked on chromosome 2.

The table shows some distances apart of the gene loci, as determined by breeding experiments.

gene loci	distance between loci / map unit
B/b and P/p	6.0
B/b and V/v	18.5
P/p and V/v	12.5
P/p and C/c	21.0
V/v and C/c	8.5

What is the correct sequence of the loci on chromosome 2?

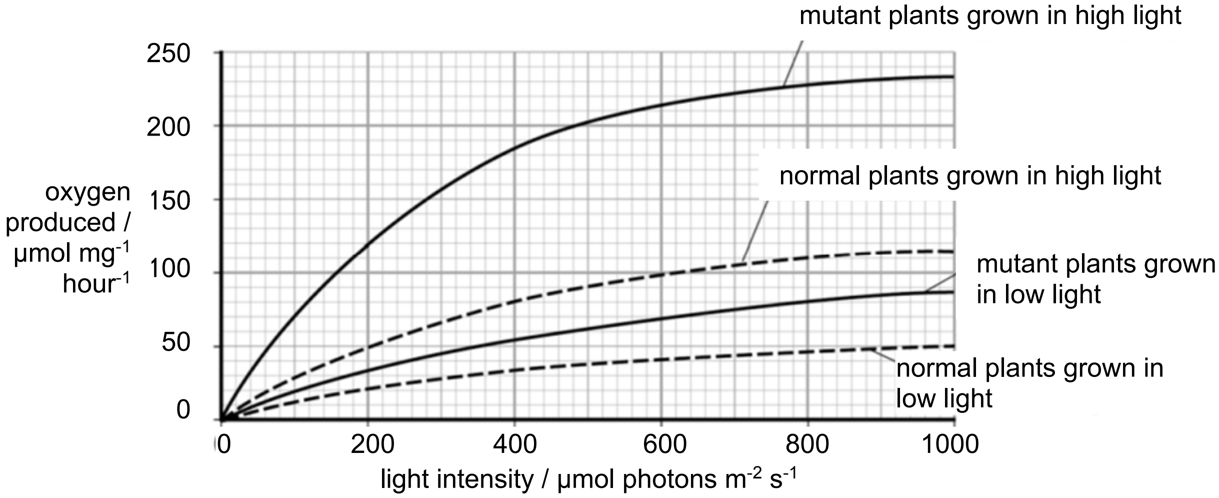
A	B/b	P/p	V/v	C/c
B	C/c	V/v	B/b	P/p
C	P/p	V/v	C/c	B/b
D	V/v	B/b	P/p	C/c

22 Chloroplasts contain chlorophyll a and chlorophyll b. Scientists found tobacco plants with a mutation that caused them to make more chlorophyll b than normal tobacco plants. They investigated the effect of this mutation on the rate of photosynthesis.

The scientists carried out the following investigation.

- They grew normal and mutant tobacco plants. They grew some of each in low light intensity and grew others in high light intensity.
- They isolated samples of chloroplasts from mature plants of both types.
- Finally, they measured oxygen production by the chloroplasts they had isolated from the plants.
- In each trial, the scientists collected oxygen for 15 minutes.

The graph shows the scientists' results.



What can be concluded by the scientists based on the results they obtained?

1 The mutant plants that produced more chlorophyll b would grow faster than normal plants in all light intensities.

2 At all light intensities, chloroplast from mutant plants have a faster production of ATP and NADPH, leading to the faster rate of light-independent reaction.

3 The difference in the oxygen produced by the chloroplasts over 15 minutes from the mutant plants grown in low and high light intensities at a light intensity of 500 μmol photons m^{-2} s^{-1} during these trials is 35 μmol O_2 mg^{-1}.

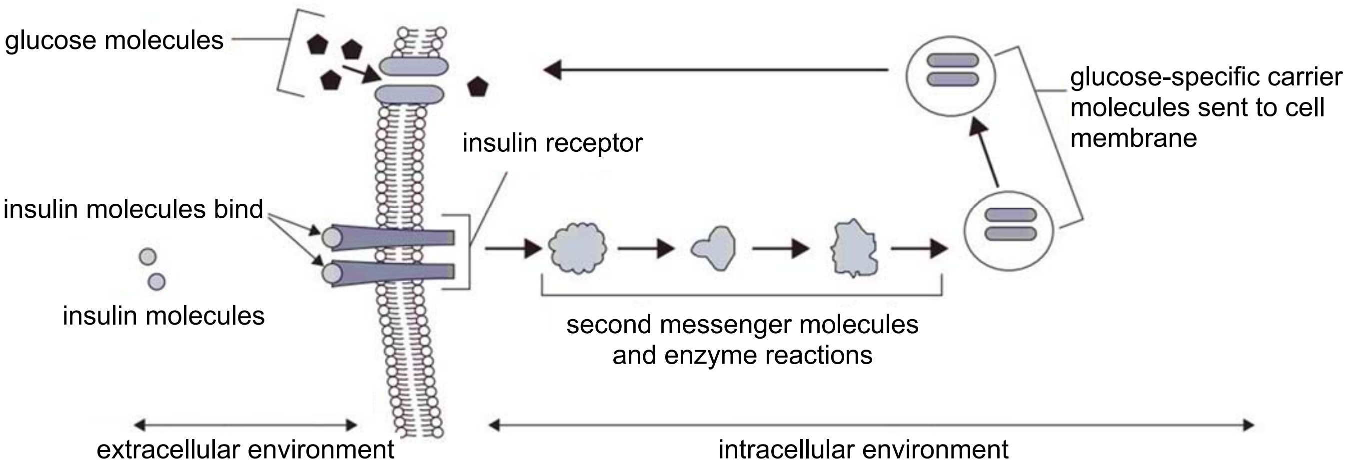
A 1, 2 and 3

B 1 and 2

C 1 and 3

D 2 and 3

23 The diagram shows a summary of the steps in an insulin signalling pathway that results in increased glucose uptake.



A scientist studied the insulin signalling pathways of two female patients, X and Y.

The results are as follows:

- patient X has a similar pathway as that shown in the diagram
- patient Y has a mutation in the gene that encodes the insulin receptor.

Which statements can be concluded from the study?

1 Insulin acts as a hydrophilic signalling molecule.
2 There would be more glucose-specific carrier molecules in patient Y's cell membranes than in patient X's.
3 The binding of insulin molecules to the receptor initiates signal transduction and the uptake of glucose into patient X's cells.
4 The presence of insulin in patient Y's cells would cause an increase in the concentration of the second messenger molecules.

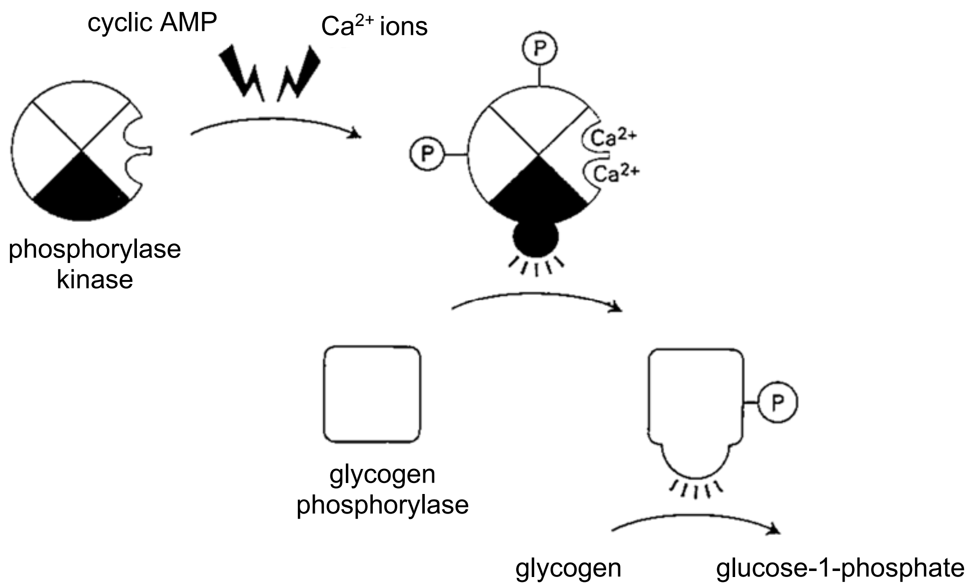
A 1, 2, 3 and 4

B 1, 2 and 4

C 1 and 3

D 3 only

24 The diagram shows how the enzyme phosphorylase kinase integrates signals from the cyclic AMP-dependent and Ca^{2+}-dependent signalling pathways.



The following statements describe the series of reactions that occur to control glycogen breakdown in liver and muscle cells.

1 Phosphorylase kinase is composed of four subunits.
2 One subunit is the protein kinase that catalyses the addition of phosphate to the glycogen phosphorylase to activate it for glycogen breakdown.
3 The other three subunits are regulatory proteins that control the activity of the catalytic subunits. Two of these three subunits contain sites for phosphorylation by PKA, which is activated by cyclic AMP.
4 The remaining subunit is calmodulin, which binds Ca^{2+} ions when its cytosolic concentration rises.
5 The regulatory subunits control the equilibrium between active and inactive conformations of the catalytic subunit.

Which statement can be concluded?

A Phosphorylase kinase has a tertiary level of protein structure.

B Glycogen phosphorylase is broken down and releases glycogen from glucose-1-phosphate molecules.

C Phosphorylase kinase adds a phosphate group to glycogen phosphorylase, inhibiting the latter.

D Binding of Ca^{2+} ions to calmodulin activates calmodulin to produce a cellular response.

25 A student recorded the shell colour and banding pattern of all empty shells of the brown-lipped snail, *Cepaea nemoralis*, found in a small deciduous wood. Wide variation in shell colour and banding pattern, which are genetically controlled, was shown. Of the shells collected, 38% were damaged and 62% were intact.

Gene **C** codes for the colour of the shell. There are six alleles.

C^{B}	C^{DP}	C^{PP}	C^{FP}	C^{DY}	C^{PY}
brown	dark pink	pale pink	faint pink	dark yellow	pale yellow

most dominant $\longrightarrow$ least dominant

Gene **B** codes for the presence or absence of bands. There are two alleles.

$\mathbf{B}^{0}$ = unbanded, dominant allele
$\mathbf{B}^{B}$ = banded, recessive allele

Which conclusion drawn from this information is valid?

A In the wood, brown, unbanded phenotypes will occur in the highest frequency and banded, pale yellow phenotypes will occur in the lowest frequency.

B More than one selection pressure is likely to be acting to maintain the polymorphism shown in shell colour and banding pattern in the wood.

C Predation by birds, such as the song thrush, is the only selection pressure acting on *C. nemoralis* in the small deciduous wood.

D The proportion of each phenotype in the population of *C. nemoralis* in the wood will be similar to the proportion of each phenotype recorded for the empty shells.

26 The rufous bristlebird, *Dasyornis broadbenti*, is a ground-dwelling songbird. The rufous bristlebird is found in gardens near thick, natural vegetation and builds nests in shrubs close to the ground. The rufous bristlebird feeds on ground-dwelling invertebrates. It is a weak flyer and is slow to go back to areas from which it has been previously eliminated.

Two distinct populations of rufous bristlebird exist in Victoria, Australia. The distribution of each population is shown on the map of Victoria. The distance between Population A and Population B is over 200 km and both of the rufous bristlebird populations in Victoria are small.

With reference to Darwin's theory of natural selection, which statements explain why the rufous bristlebird is at risk of extinction?

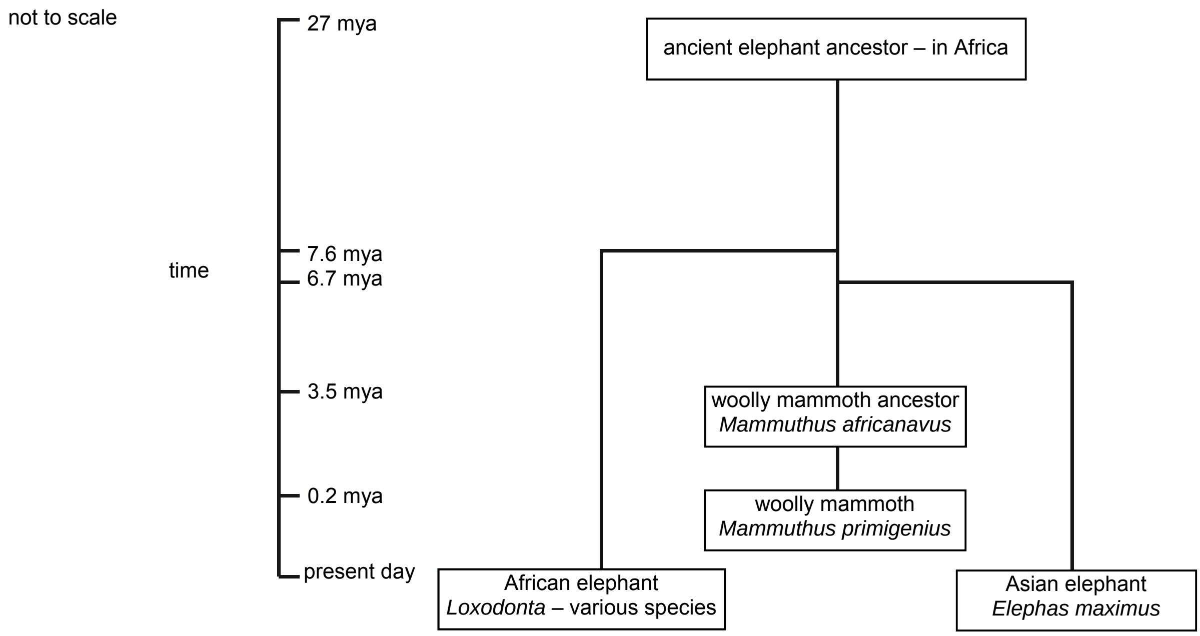
1 There is reduced genetic variation in the limited gene pool, thus reducing the chances of survival.
2 There is insufficient variation in the population to survive within their current environment.
3 Inbreeding could result in an increased chance of genetic diseases.
4 Changes in allele and genotype frequencies is random due to genetic drift.
5 Being ground-dwelling, the eggs of the rufous bristlebirds may be frequently eaten.

A 1, 2, 3, 4 and 5 **B** 1, 2, 4 and 5 **C** 1, 2 and 3 **D** 3, 4 and 5

27 The extinct woolly mammoth, *Mammuthus primigenius*, was closely related and similar in size to present-day elephants. It was herbivorous and highly adapted to very cold climates, with a dense undercoat and small, fur-lined ears. The table shows information about woolly mammoths inferred from extensive fossil evidence.

number of years ago	event	abundance of mammoths
200 000	woolly mammoths first appeared in north-east Siberia and later spread to Europe, Asia and North America	very high
120 000 – 100 000	climate warmed	moderate
100 000	climate cooled again	
15 000	first evidence of human hunting where a spear was found in a mammoth vertebrae	low
14 000 – 10 000	mammoths extinct in North America, Europe and most of Northern Asia	very low, found only in Siberia
3900	last woolly mammoth died on an island off the coast of Siberia	non-existent

It has been suggested that elephant-like ancestors of the woolly mammoth left Africa 3.5 million years ago (mya) and lived in Central Europe. The diagram shows a summarised phylogenetic tree based on mitochondrial and chromosomal DNA from fossils and living elephants.

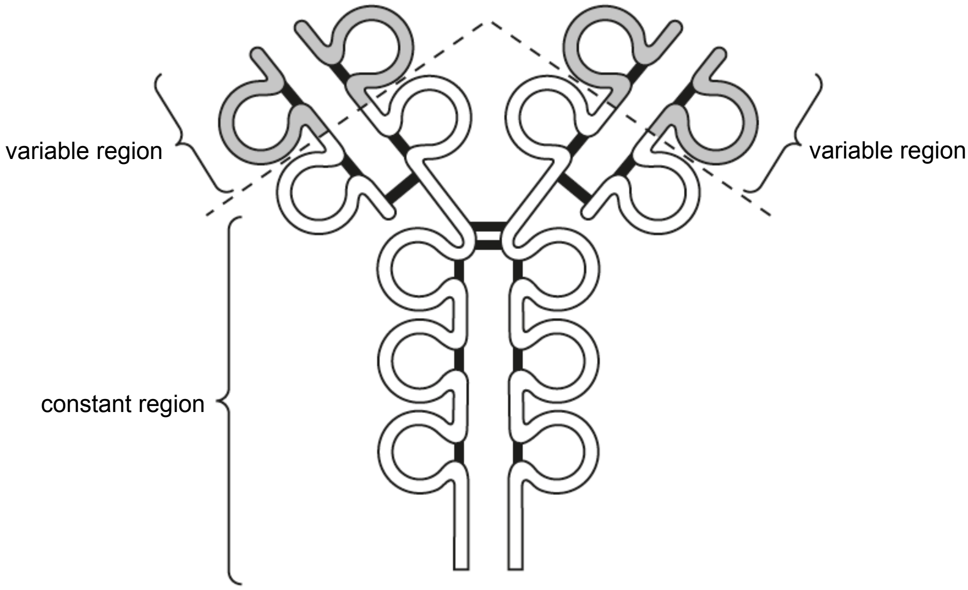


Which statements correctly describe the evolution of elephants?

1 Allopatric speciation occurred when *M. africanavus* left Africa and characteristics of *M. primigenius* changed through speciation due to different environments.
2 Within the species *M. africanavus*, genetic differences accumulated and variation led to adaptive radiation.
3 *M. primigenius* had limited genetic variation and due to increased hunting by humans, was unable to evolve.
4 *E. maximus* lives in an environment with more stable selection pressures compared to *M. primigenius*.

A 1, 2 and 4 **B** 1, 3 and 4 **C** 1 and 3 **D** 3 and 4

28 The diagram shows the structure of an antibody molecule.



How many of the statements are correct?

1. The two variable regions have different amino acid sequences from each other, which increase the diversity of antigens that the antibody can bind to.

2. Only the light chains form the antigen-binding sites.

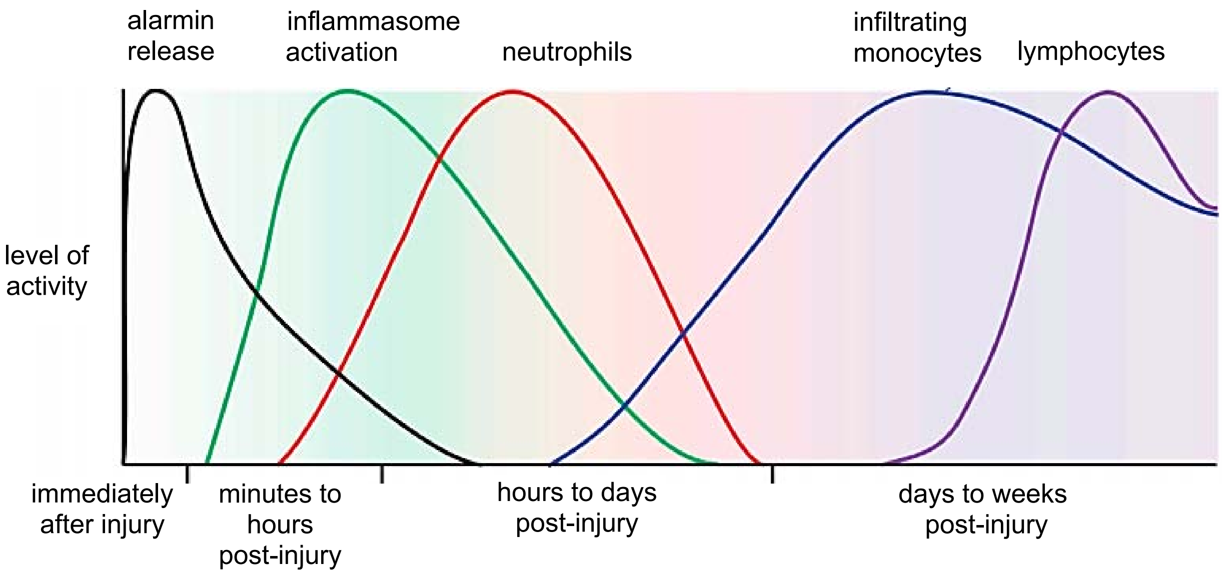
3. The hypervariable region within each variable region provide the structural basis for the diversity of the binding sites.

4. The antibody can have multiple epitopes at the variable region to allow the antibody to bind to many types of antigen.

A one **B** two **C** three **D** four

29 The graph shows the kinetics of the molecular and cellular immune responses to a brain injury.

- Alarmins, also known as danger signals, are released rapidly in the extracellular fluid during infection and tissue damage.
- Levels of inflammasome are elevated soon, and when activated, triggers production of cytokines.
- Neutrophils arrive hours after injury and stay for several days, while monocytes begin infiltrating within the first day and remain present.
- Lymphocytes begin to arrive days to weeks post-injury.



Which statement is **not** true?

A Neutrophils are the first cells recruited to the wound site in order to ingest, kill and digest pathogens.

B The pathogens secrete cytokines via inflammasomes to attract passing neutrophils to the injury site, resulting in the peak in neutrophil levels.

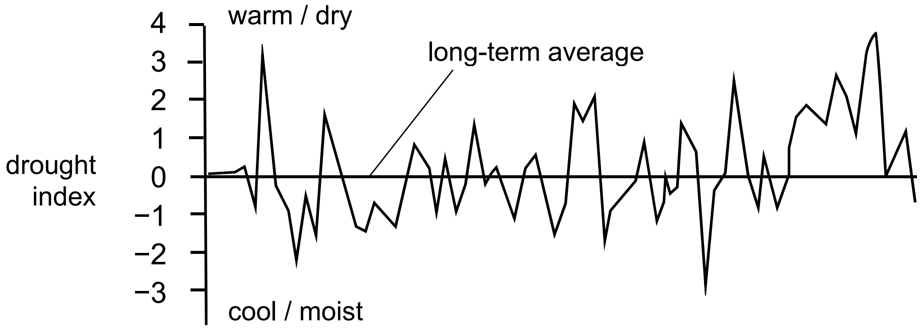
C Monocytes enter the extravascular tissues and differentiate into macrophages, which unlike neutrophil, survive in the injury site for long periods.

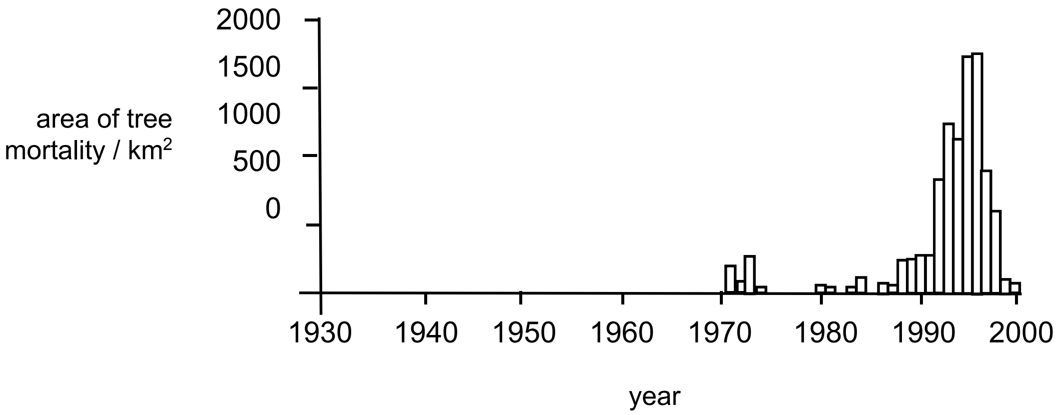
D Lymphocytes undergo clonal selection and clonal expansion, and remained in high levels post-injury due to the retention of memory cells.

30 With climate change, some regions on Earth are experiencing warmer temperatures. Warmer temperatures favour some species of pest, for example the spruce beetle, *Dendroctonus rufipennis*.

Since the first major pest outbreak, the spruce beetles have severely destroyed approximately 400,000 hectares of trees in Alaska and the Canadian Yukon.

The graphs show the drought index, a combination of temperatures and precipitation and the area of spruce trees destroyed annually.





Which statement best explains the trends observed in the graphs?

A From the late 1980s, prolonged period of drought and warm weather led to tree mortality because warmer climate resulted in an increased number of generations of the spruce beetles.

B Between 1930 and 1970, there was no tree mortality because the spruce beetles were had other food sources.

C The pest outbreak occurred because wet, cool climate caused the spruce beetles to be more active and thus reproduce at a faster rate.

D The spike in tree mortality was likely not related to climate, but due to a random catastrophic event that wiped out an entire patch of forest.

--END OF PAPER---

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